# Numerical Methods in Astrophysics (0321488101): Numerov Method for Bound States in the One-Dimensional Schrödinger Equation

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## Contents

# 1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with parity-aware shooting. The aim is to validate the method on problems with known analytic spectra, quantify its numerical accuracy, and then apply it to more interesting bound-state potentials such as a double well and a finite square well. Scattering from finite barriers is included as a secondary extension.

This is a boundary-value eigenvalue problem rather than a standard initial-value problem: only special energies satisfy the required boundary behavior. Numerov is well suited to this task because the equation has the second-order form $\psi'' = q(x)\psi$ with no first-derivative term. The project therefore emphasizes both the computed spectra and the numerical checks behind them, including analytic benchmarks, convergence studies, parity and node structure, and probability conservation in the scattering extension.

The full Python implementation is available in this repository. The accompanying `README.md` and documentation describe how to run the code and reproduce the results.

# 2 Computational approach

## 2.1 The Schrödinger equation and Numerov method

In dimensionless units, the Schrödinger equation is written as:

$$\psi''(x) = 2\,[V(x) - E]\psi(x) \tag{1}$$

The wavefunction is integrated with the Numerov scheme on a uniform grid. For the symmetric bound-state potentials studied here, $V(-x) = V(x)$, so the eigenstates can be chosen to have definite parity: even states satisfy $\psi'(0) = 0$ and odd states satisfy $\psi(0) = 0$. This symmetry lets the solver work only on the half-domain $[0, x_{\max}]$ and reconstruct the other half by reflection. Eigenvalues are then obtained by searching for zeros of a scalar mismatch function and refining the resulting brackets with robust root finding. The implementation follows the standard Numerov-shooting strategy described in [1].

## 2.2 Shooting and root finding

The boundary-value problem is reformulated as a root-finding problem for a mismatch function $M(E)$. For a trial energy $E$, the Schrödinger equation is integrated and a boundary mismatch is evaluated. The code uses two related formulations. For the infinite square well, finite square well, and quartic double well, the solution is shot outward from $x = 0$ and the mismatch is the boundary value $M(E) = \psi_E(x_{\max})$. For the harmonic oscillator and the RK4 comparison, the integration is instead started from the decaying tail at $x_{\max}$ and the parity condition is enforced at the origin:

$$\text{even states:} \quad M(E) = \psi'_E(0), \qquad \text{odd states:} \quad M(E) = \psi_E(0) \tag{2}$$

The inward formulation is specialized to smooth confining problems where the truncation point already lies in a forbidden decaying region. In all cases, eigenvalues correspond to roots of $M(E) = 0$, found by bracketing sign changes and then refining them numerically. The outward solver uses bisection followed by a short safeguarded polishing step inside the final bracket, while the inward and RK4 solvers use straight bisection. In the inward even-state case, $\psi'(0)$ is evaluated with a fourth-order backward stencil so the boundary condition does not reduce the overall observed order of accuracy.

## 3 Results

Each potential is used to validate, test, or demonstrate different aspects of the numerical method, ranging from analytic benchmarks to nontrivial quantum behavior.

### 3.1 Infinite square well

The infinite square well serves as a benchmark problem with known analytic spectrum:

$$E_n = \frac{(n+1)^2\pi^2}{8a^2} \tag{3}$$

The potential is:

$$V(x) = \begin{cases} 0 & , |x| \leq a \\ \infty & , |x| > a \end{cases} \qquad \text{(implemented numerically with a large finite } V_0\text{)} \tag{4}$$

In the production calculation, the well half-width is $a = 1$, the outward-shooting half-domain uses $x_{\max} = a = 1$, the Numerov grid uses $n_{\text{grid}} = 2500$ points, and the numerical wall height is set to $V_0 = 10^6$. A separate grid-refinement study is used to check convergence at fixed $x_{\max} = 1$.

The numerical eigenvalues agree with the analytic values to essentially machine precision, as shown in Fig. 1; in the saved run the first four relative errors are of order $10^{-11}$–$10^{-10}$. The probability densities and state plots in Figs. 2 and 6 show the expected parity and node structure, while the mismatch scans in Fig. 3 isolate the first four roots cleanly. The convergence study in Fig. 5 gives fitted exponents $p = 3.94$, 4.00, 3.98, and 4.00, so this benchmark also confirms the expected fourth-order accuracy.

### 3.2 Harmonic oscillator

#### 3.2.1 Numerov method on unbounded domains

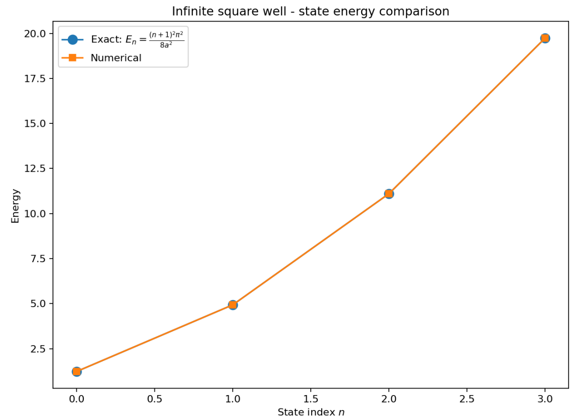
For the harmonic oscillator with $V(x) = \frac{1}{2}\omega^2 x^2$, the exact spectrum is:

$$E_n = \omega\left(n + \tfrac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{5}$$

This provides a second analytic benchmark, but on an unbounded domain. Numerically, the domain is truncated to $[-x_{\max}, x_{\max}]$, and decaying boundary conditions are imposed. In the production Numerov run, the parameters are $\omega = 1$, $x_{\max} = 8$, and $n_{\text{grid}} = 2500$. Separate studies vary the grid spacing at fixed $x_{\max}$ and the box size at approximately fixed spacing.

The numerical eigenvalues agree with the exact linear spectrum to extremely high precision, with first-four-state relative errors of order $10^{-12}$–$10^{-11}$ in the saved run. The densities and state plots in Figs. 8 and 13 show the expected parity, node structure, and growing spatial extent with energy. The mismatch scans in Figs. 9 and 10 show clean alternating even and odd roots despite the large dynamic range introduced by inward shooting from the decaying tail.

The convergence study versus grid spacing in Fig. 11 is consistent with fourth-order behavior. The saved fits give $p \approx 5.30$, 3.97, 4.57, and 4.00 for the first four states; the values above four occur only once the errors are already near the numerical floor and are therefore not interpreted as true superconvergence. The box-size study in Fig. 12 shows the expected truncation-to-discretization crossover: enlarging the domain rapidly reduces the error at first, after which the curves flatten once truncation error is no longer dominant.



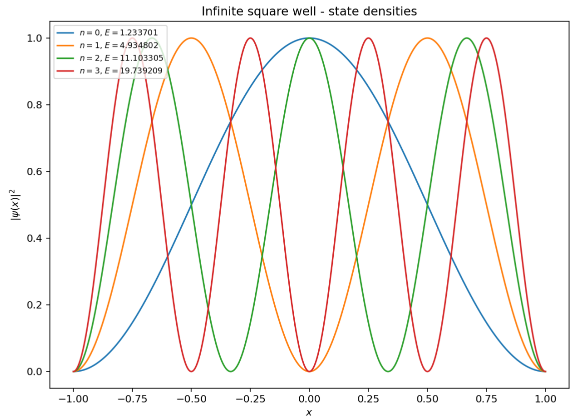
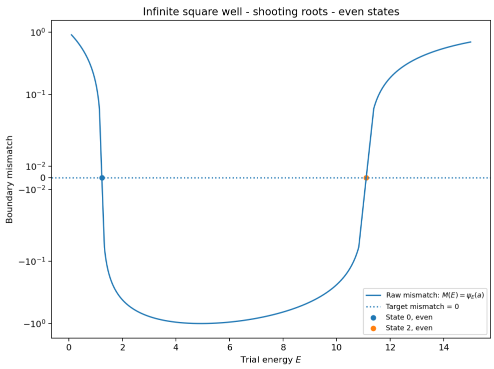




**Figure 1:** Comparison between numerical and exact energy eigenvalues for the infinite square well. The agreement is excellent for the first four states.

**Figure 2:** Probability densities $|\psi(x)|^2$ for the first four eigenstates of the infinite square well. The correct number of nodes and boundary behavior are clearly reproduced.

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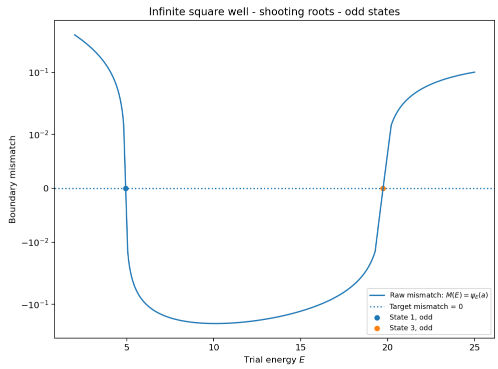
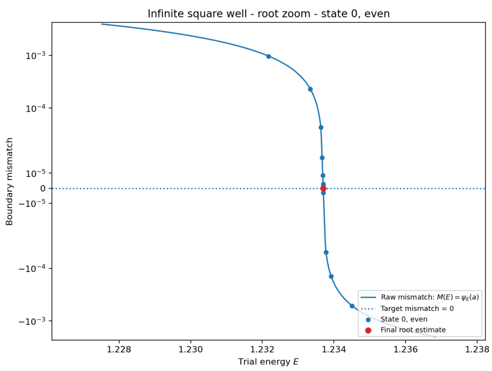



**(b)**

**Figure 3:** Raw boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues found by the shooting method.

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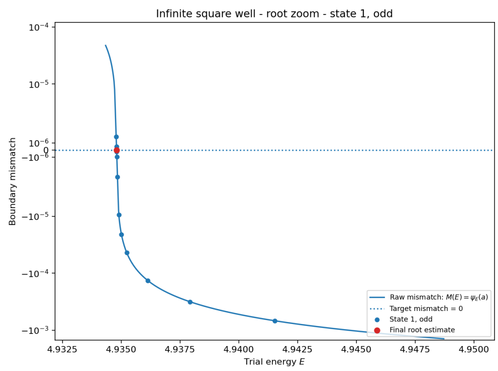


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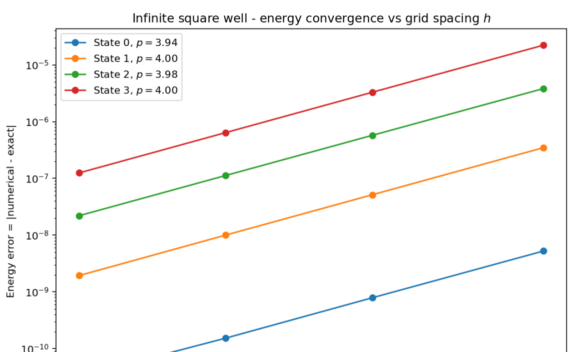
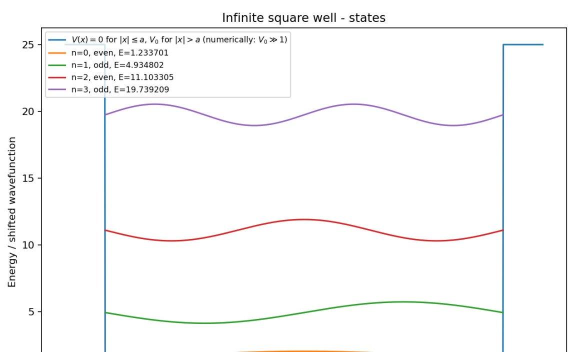
**Figure 4:** Zoomed root brackets for the lowest even (a) and odd (b) infinite-well states. These panels show the raw mismatch only inside one initial sign-changing bracket, together with the recorded bisection iterates.



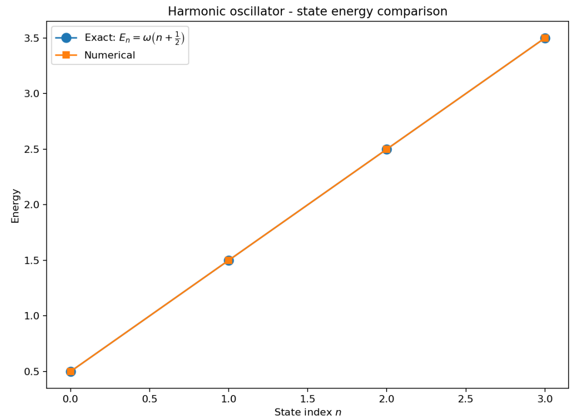



**Figure 7:** Comparison between numerical and exact energy eigenvalues for the harmonic oscillator. The agreement is excellent for the first four states.

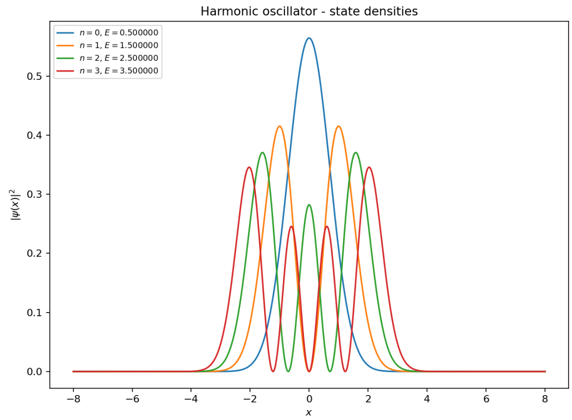
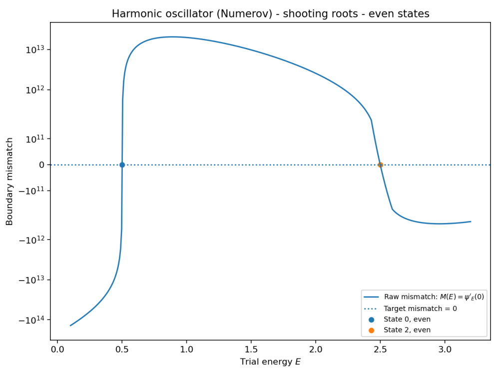
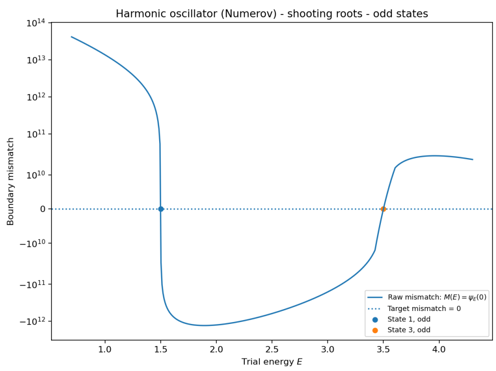



**Figure 8:** Probability densities $|\psi(x)|^2$ for the first four harmonic oscillator eigenstates. The correct node structure is clearly reproduced.

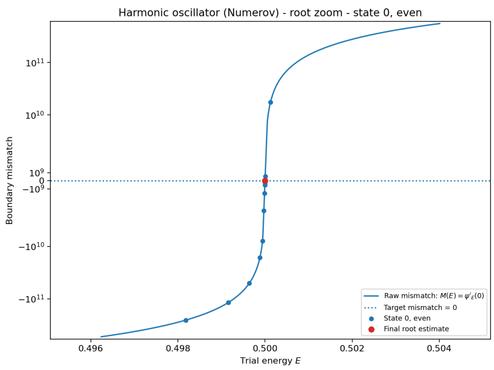



**(a)**

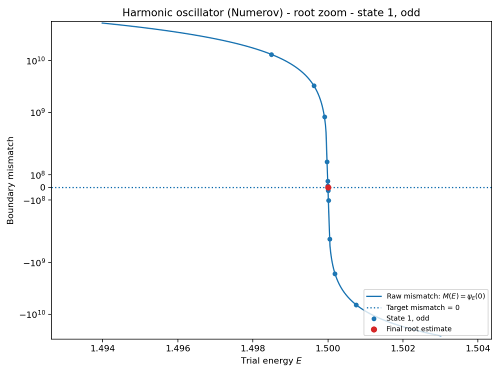



**(b)**

**Figure 9:** Raw boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues.




**(a)**




**(b)**

**Figure 10:** Zoomed root brackets for the lowest even (a) and odd (b) harmonic-oscillator states. These local views remove the global dynamic-range compression and show the raw mismatch together with the bisection iterates inside each initial bracket.

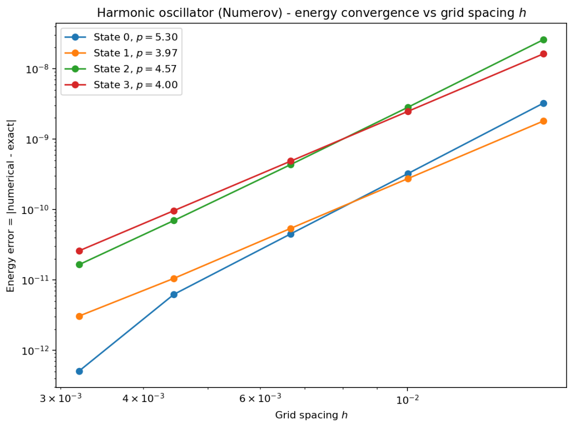



**Figure 11:** Convergence of the energy error as a function of grid spacing $h$. The slopes are close to fourth order, as expected for the Numerov method.

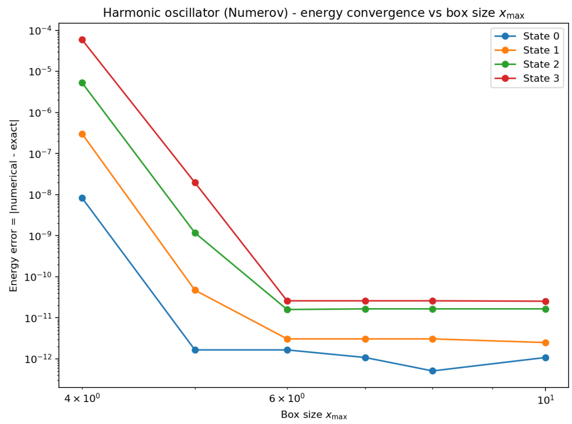



**Figure 12:** Dependence of the energy error on the domain size $x_{\max}$. Errors decrease rapidly before reaching a plateau set by discretization and floating-point limits.

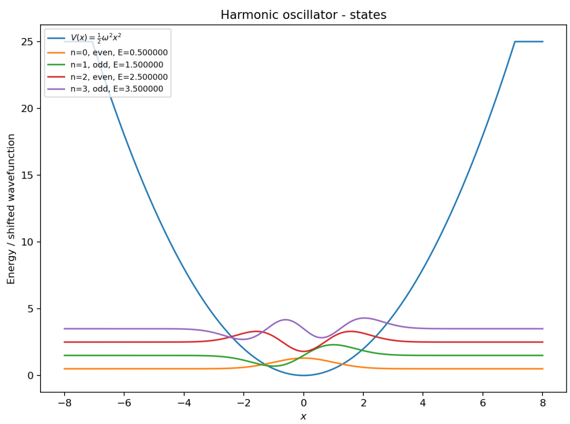



**Figure 13:** Wavefunctions for the first four harmonic oscillator states plotted together with the potential.

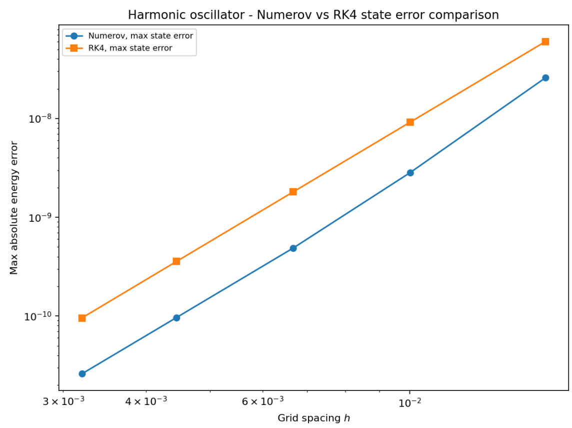



**Figure 14:** Comparison of Numerov and RK4 shooting for the harmonic oscillator. Both methods show fourth-order convergence, while Numerov gives a smaller maximum energy error at the same grid spacing.

### 3.2.2 Numerov versus Runge–Kutta

As an additional method comparison, the harmonic oscillator was also solved using a fourth-order Runge–Kutta (RK4) shooting method. In this formulation the second-order Schrödinger equation is rewritten as the first-order system:

$$\psi' = \phi, \qquad \phi' = 2[V(x) - E]\psi \tag{6}$$

This makes RK4 a useful general-purpose comparison for Numerov, which instead exploits the special second-order structure of the Schrödinger equation directly. The RK4 comparison uses the same physical parameters $\omega = 1$ and $x_{\max} = 8$ as the Numerov run, and it reuses the same family of grid spacings derived from the Numerov convergence study so the two methods are compared on matched meshes.

Figure 14 compares the maximum energy error over the computed harmonic-oscillator states as a function of grid spacing. Both methods show approximately fourth-order convergence, but Numerov gives the smaller error on the matched grids, typically by a factor of about 2–4. The RK4 slopes are also more uniformly close to four, while the Numerov fit is affected more strongly once the errors approach the numerical floor. This comparison is included mainly to show the benefit of using a method specialized to second-order equations when the boundary diagnostics are implemented with comparable care.

## 3.3 Double-well potential

The quartic double-well potential is used as the main nontrivial bound-state example. In the shifted form used here:

$$V(x) = ax^4 - bx^2 - V_{\min}, \qquad V_{\min} = -\frac{b^2}{4a} \tag{7}$$

the two minima are shifted to zero. This shift is applied with the analytic minimum rather than the sampled grid minimum $\min_i V(x_i)$ taken over the current mesh points, because the sampled minimum drifts slightly as the grid changes and would otherwise contaminate convergence studies. The analytic shift also gives a cleaner energy reference: the well bottoms sit at zero, so the reported low-lying energies can be read directly relative to the bottoms of the wells, while the potential shape and tunneling splittings remain unchanged. Unlike the infinite square well and harmonic oscillator, this system does not have a simple analytic spectrum, so the validation relies on internal consistency checks, convergence studies, parity structure, and physically expected tunneling behavior. The production run uses $a = 1$, $b = 6$, the analytic minimum shift $V_{\min} = b^2/(4a)$, a box size $x_{\max} = 4$, and $n_{\text{grid}} = 4000$. Separate studies are used to test grid refinement, box-size sensitivity, and the dependence of the lowest splitting on the parameter $b$.

Figure 15 shows the first four computed states together with the double-well potential. The lowest two form a nearly degenerate even-odd pair, $E_0 \approx 2.357273$ and $E_1 \approx 2.359372$, with a splitting of about $2.10 \times 10^{-3}$; the next pair, $E_2 \approx 6.548814$ and $E_3 \approx 6.684430$, is more strongly split. The densities in Fig. 16 show localization near the two wells, while the root diagnostics in Figs. 17 and 18 show clean parity-separated roots.

Since no closed-form spectrum is available, grid convergence is measured through successive refinements at fixed domain size using the surrogate error $|E(h_i) - E(h_{i+1})|$. Figure 19 still shows stable high-order behavior, although fits above fourth order appear once the differences approach the numerical floor. The box-size study in Fig. 20 is especially important for this problem: it shows that small boxes visibly contaminate the low states, while $x_{\max} \approx 4$ is large enough for the remaining error to be negligible relative to a larger-box reference. Finally, Fig. 21 shows the main physical trend of the section: as $b$ increases from 3 to 8, the lowest even-odd splitting falls from about $3.52 \times 10^{-1}$ to $1.59 \times 10^{-5}$, indicating tunneling suppression as the wells deepen and separate.

### 3.4 Finite square well

The finite square well is included as an additional nontrivial bound-state test. Unlike the infinite square well, the potential walls are finite, so bound states can leak into the classically forbidden region. The potential used here has the form:

$$V(x) = \begin{cases} 0 & , |x| \leq a \\ V_0 & , |x| > a \end{cases} \tag{8}$$

and only states with $E < V_0$ are true bound states. In the production run, the well parameters are $a = 1$ and $V_0 = 12$, while the outward-shooting box uses $x_{\max} = 4$ and the Numerov mesh uses $n_{\text{grid}} = 3000$. A supplementary grid-refinement check compares coarser solutions to the finest saved run, so the plotted quantity is $|E_N - E_{\text{ref}}|$ with the $n_{\text{grid}} = 3000$ spectrum acting as the practical reference. Unlike the smooth benchmark potentials, however, the finite square well is only piecewise smooth: $V(x)$ jumps at $|x| = a$. That discontinuity degrades the observed asymptotic convergence order, so this refinement study should be read mainly as a practical convergence check rather than a clean fourth-order Numerov benchmark.

Figure 22 shows the first four computed states together with the finite well potential. For $V_0 = 12$ and $a = 1$, the first three states are well localized inside the well, while the fourth lies close to the barrier top and shows much stronger leakage into the exterior region. The densities in Fig. 23 make this contrast especially clear.

The finite-well root diagnostics in Figs. 24 and 25 also clarify the near-threshold behavior. As $E \to V_0^-$, the exterior decay constant tends to zero, so the raw mismatch can become small without creating an additional bound-state root; the zoomed brackets show that the reported high-energy states still correspond to genuine sign changes. The grid-refinement study is included mainly as a practical convergence check, since the discontinuity at the well edge lowers the observed asymptotic order relative to the smooth benchmark cases.

### 3.5 Scattering and resonant tunneling

#### 3.5.1 Single barrier potential

As a secondary extension beyond bound states, the Numerov framework is also applied to one-dimensional scattering from a finite rectangular barrier. For each incident energy $E$, the complex wavefunction is propagated through the barrier region and decomposed into incident, reflected, and transmitted waves. This gives the transmission and reflection probabilities $T(E)$ and $R(E)$. The integration is started from the right free region, not the left, because the right boundary condition is the simplest one to impose directly: for a left-incident scattering state, the far-right asymptotic solution contains only the transmitted outgoing wave $e^{ikx}$. The left asymptotic region is more complicated because it contains a superposition of incident and reflected waves, so those amplitudes are recovered afterward by decomposing the numerically propagated left solution.

The shared scattering grid for both barrier problems is $x \in [-8, 8]$ with $n_{\text{grid}} = 4000$ points. For the single-barrier run, the barrier parameters are $V_0 = 5$, width 1.2, and center 0, while the incident energy is scanned over a range that covers both below-barrier and above-barrier behavior.

The single-barrier result is shown in Fig. 26. Transmission is strongly suppressed at low incident energy and increases as the energy rises through and above the barrier. Even for $E > V_0$,

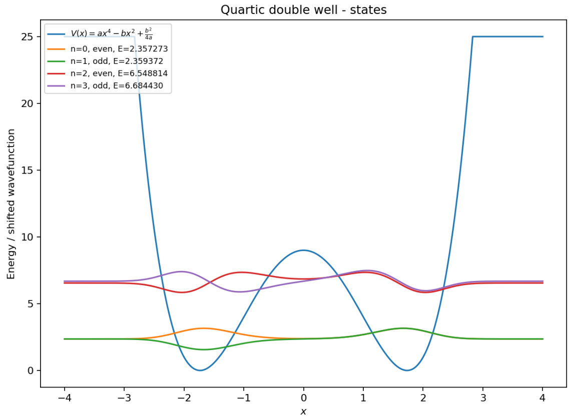



**Figure 15:** First four bound states of the quartic double-well potential. The lowest two states form a nearly degenerate tunneling doublet.

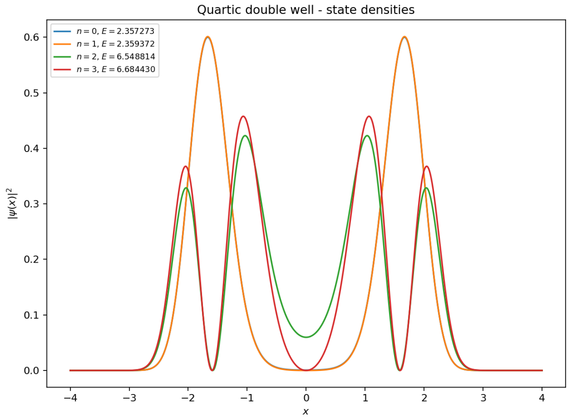
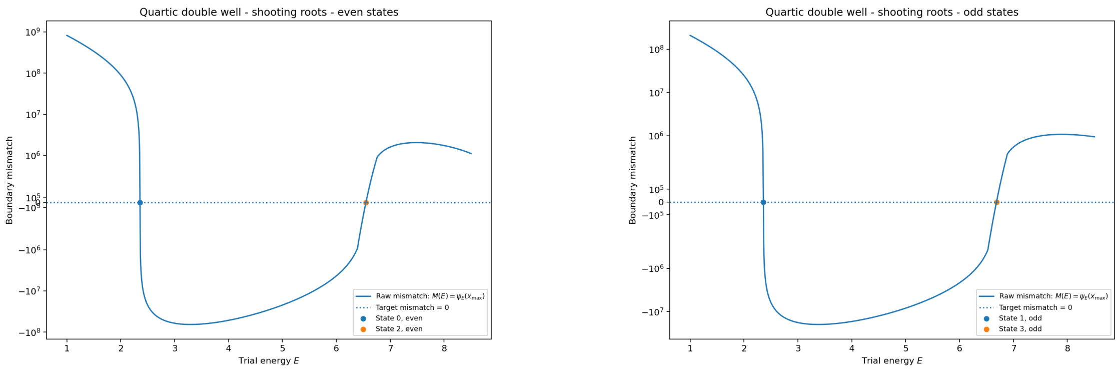
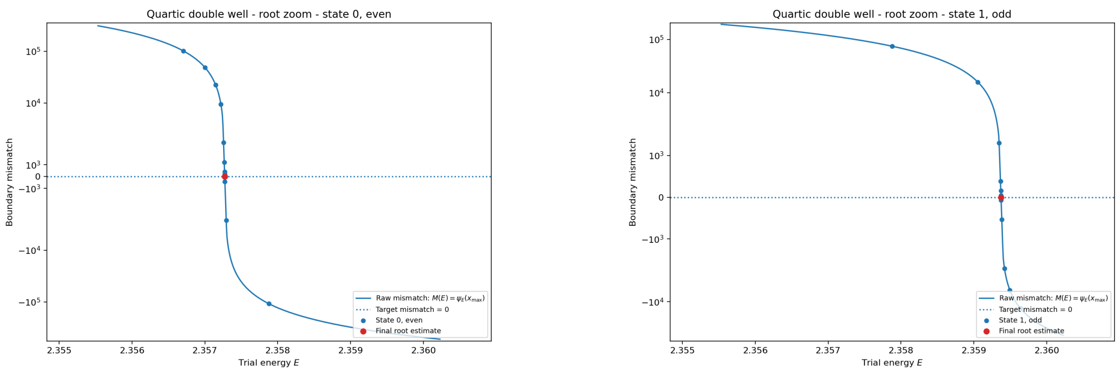



**Figure 16:** Probability densities $|\psi(x)|^2$ for the first four double-well states. The lowest even and odd states have nearly identical densities, as expected for a tunneling doublet.




**(a)** **(b)**

**Figure 17:** Raw shooting mismatch functions for even (top) and odd (bottom) double-well states. The zero crossings define the numerical eigenvalues.




**(a)** **(b)**

**Figure 18:** Zoomed root brackets for the lowest even (a) and odd (b) double-well states. These local panels show the raw mismatch and the recorded bisection sequence inside the initial sign-changing brackets.

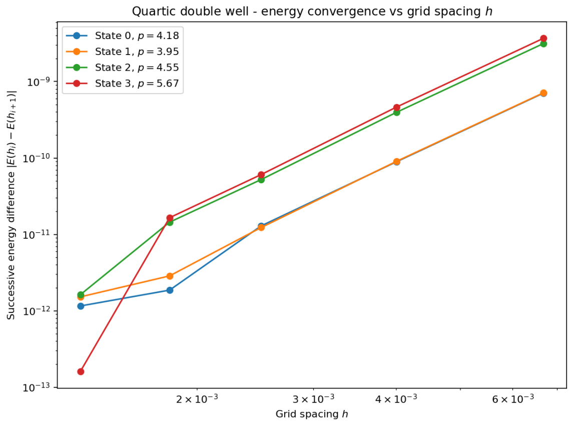



**Figure 19:** Grid convergence for the quartic double well. Since no exact spectrum is available, the vertical axis shows the successive-refinement surrogate $|E(h_i) - E(h_{i+1})|$ rather than $|E_{\text{numerical}} - E_{\text{exact}}|$. The curves still decrease with clear high-order behavior under grid refinement.

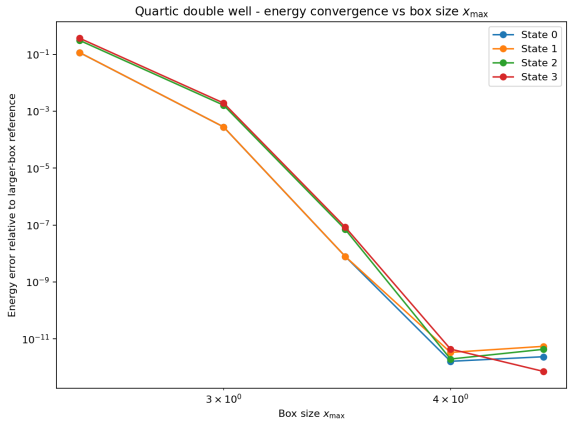



**Figure 20:** Dependence of the double-well energy errors on box size, measured relative to a separate larger-box reference calculation at approximately fixed grid spacing. The errors decrease rapidly as the domain is enlarged and then reach a numerical plateau.

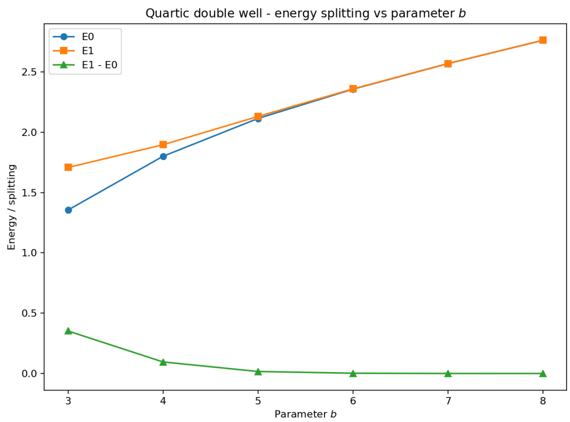



**Figure 21:** Lowest two double-well energies together with their splitting as the parameter $b$ is varied. The splitting decreases as the barrier becomes stronger, consistent with tunneling suppression.

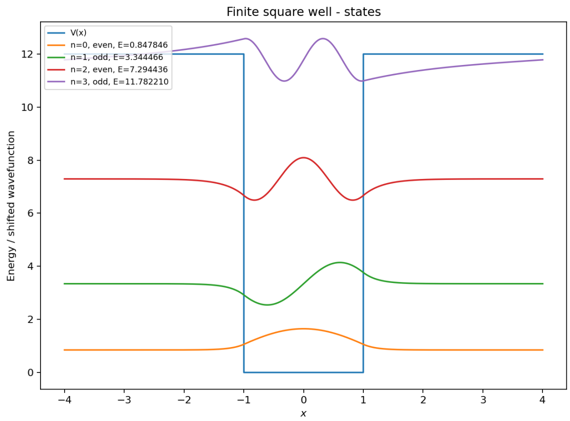



**Figure 22:** First four finite square well states plotted together with the finite barrier potential. States below the barrier are localized in the well, while the near-threshold state has a larger evanescent tail.

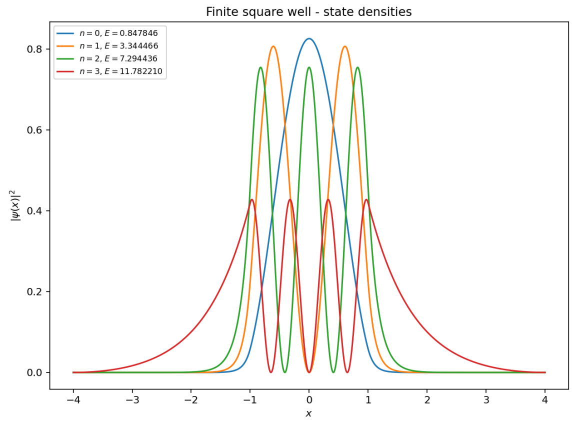
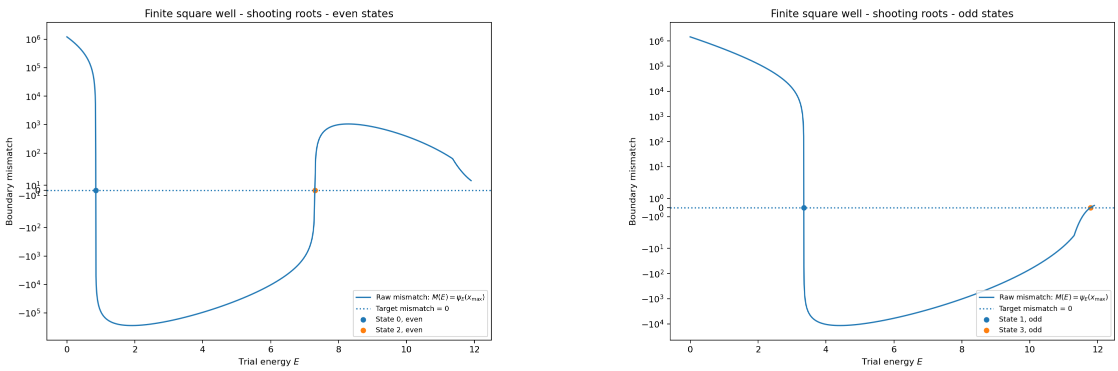
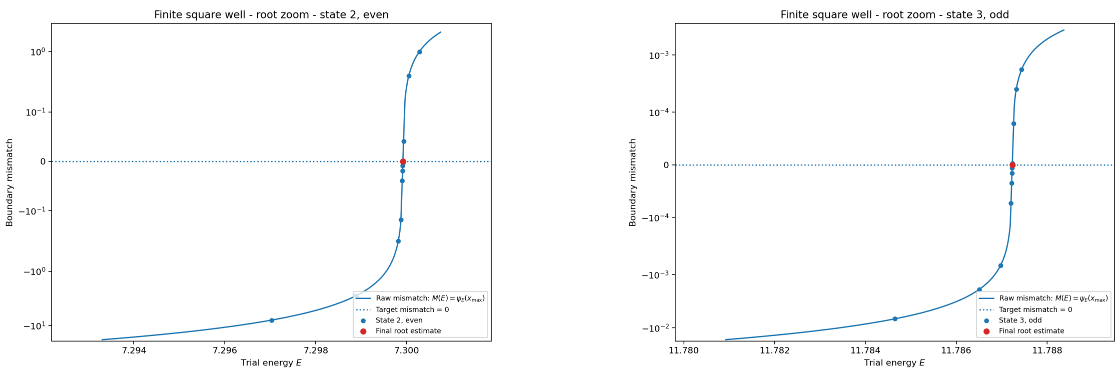



**Figure 23:** Probability densities $|\psi(x)|^2$ for the first four finite square well states. The densities show both interior oscillation and exterior exponential leakage.




**(a)** **(b)**

**Figure 24:** Raw shooting mismatch functions for even (a) and odd (b) finite-well states. The global scans include the near-threshold regime just below the barrier top $V_0$.




**(a)** **(b)**

**Figure 25:** Zoomed near-threshold brackets for the higher even (a) and odd (b) finite-well bound states. These views isolate the genuine sign-changing roots close to the barrier top and show the full recorded bisection history.

transmission is not exactly one because the finite interfaces still cause partial reflection. The conservation check $T(E)+R(E)$ remains numerically indistinguishable from one across the saved scan.

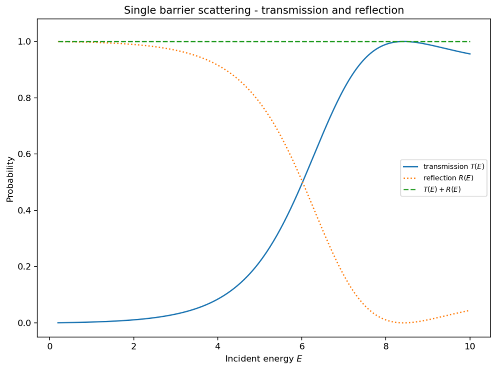



**Figure 26:** Transmission and reflection probabilities for scattering from a finite rectangular barrier. At low energies reflection dominates, while at high energies transmission approaches unity. The conservation check $T(E) + R(E)$ remains close to one.

### 3.5.2 Double barrier potential

The double-barrier potential is used to demonstrate resonant tunneling. In this case the two barriers create an intermediate well, so certain incident energies can form quasi-bound states between the barriers. At these energies, destructive reflection and constructive transmission lead to sharp peaks in the transmission coefficient.

For the production double-barrier run, each barrier has height $V_0 = 5$, the barrier width is 0.6, the central well width is 1.4, the structure is centered at $x = 0$, and the incident energy is scanned across the resonance region below the barrier top. Resonance candidates are then identified from peaks in the transmission curve.

Figure 27 shows two clear resonances in the saved transmission curve: a narrow low-energy peak near $E \approx 1.13938$ and a broader high-energy peak near $E \approx 4.35847$, both with transmission very close to one. The conservation check $T(E) + R(E)$ remains essentially exact across the scan. Figure 28 shows the probability density at the first resonance, where the wavefunction is strongly enhanced in the region between the barriers, identifying the quasi-bound state responsible for resonant tunneling.

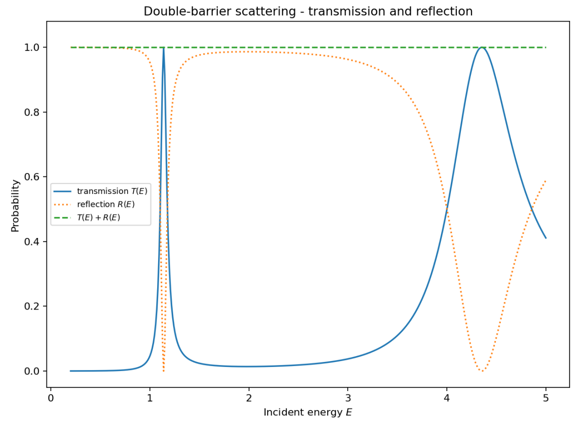



**Figure 27:** Transmission and reflection probabilities for the double-barrier potential. Sharp transmission peaks occur at resonant energies, where reflection is suppressed and $T(E) + R(E)$ remains close to one.

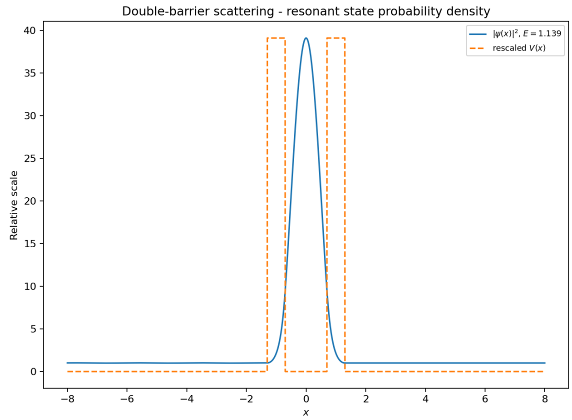



**Figure 28:** Probability density at the first resonant energy. The wavefunction is strongly enhanced between the two barriers, showing the quasi-bound state responsible for resonant tunneling.

# 4 Discussion

The calculation was checked in four complementary ways: automated tests of key numerical components, analytic benchmark spectra for the infinite square well and harmonic oscillator, inspection of parity and node structure in the wavefunctions, and convergence studies with respect to both grid spacing and domain size. Taken together, these checks show that the method is reliable when the boundary treatment is handled with the same care as the integration formula itself.

The convergence studies also identify the main practical limitations. In the smooth benchmark problems, the observed behavior is consistent with fourth-order accuracy, while slopes slightly above four occur only once the errors are already near the numerical floor. The harmonic-oscillator box-size study separates domain truncation from discretization error, and the RK4 comparison shows that method order alone is not enough: once the inward boundary derivative is evaluated accurately, Numerov gives smaller errors than RK4 at matched grid spacing. For the quartic double well, where no exact spectrum is available, correctness is established through self-consistency, parity-separated diagnostics, and stable refinement behavior. The finite well and scattering extensions show that the same framework also handles barrier penetration and resonant tunneling without losing flux conservation or qualitative physical consistency.

# 5 Conclusion

This project implemented a Numerov shooting solver for the one-dimensional time-independent Schrödinger equation and validated it on analytic benchmark problems. The method recovers the infinite-well and harmonic-oscillator spectra to very high precision and shows the expected high-order convergence when the boundary treatment is implemented consistently.

The same framework also resolves nontrivial physics beyond the benchmarks, including tunneling splitting in a quartic double well, barrier penetration in a finite square well, and resonant transmission through double barriers. Overall, the results show that Numerov shooting is an accurate and flexible method for one-dimensional quantum calculations when boundary diagnostics and convergence checks are treated as part of the numerical method itself.

## References

[1] Tao Pang. *An Introduction to Computational Physics*. 2nd ed. Cambridge University Press, 2006.